

NETWORKING FOR ETHICAL HACKERS

◆ 1. What is a Network?

A **network** is a collection of interconnected devices that communicate using protocols like **TCP/IP**.

Types of networks:

- **LAN** – Local internal networks
 - **WAN** – Large or global networks
 - **MAN, WLAN, PAN** – Specialized forms
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◆ 2. What is the Internet?

The **Internet** is a global “network of networks” linked using the **TCP/IP suite**, enabling communication worldwide.

◆ 3. Networking Basics

Concept	Description
IP Address	Unique address for each device on a network.
MAC Address	Hardware identifier for network interfaces.
Subnet	Divides networks into smaller, manageable parts.
Gateway	Device that connects your network to other networks.
DNS (Domain Name System)	Translates domain names into IP addresses.
DHCP (Dynamic Host Configuration Protocol)	Automatically assigns IP addresses to devices.
Protocols (TCP, UDP, HTTP, etc.)	Rules that devices follow to communicate.

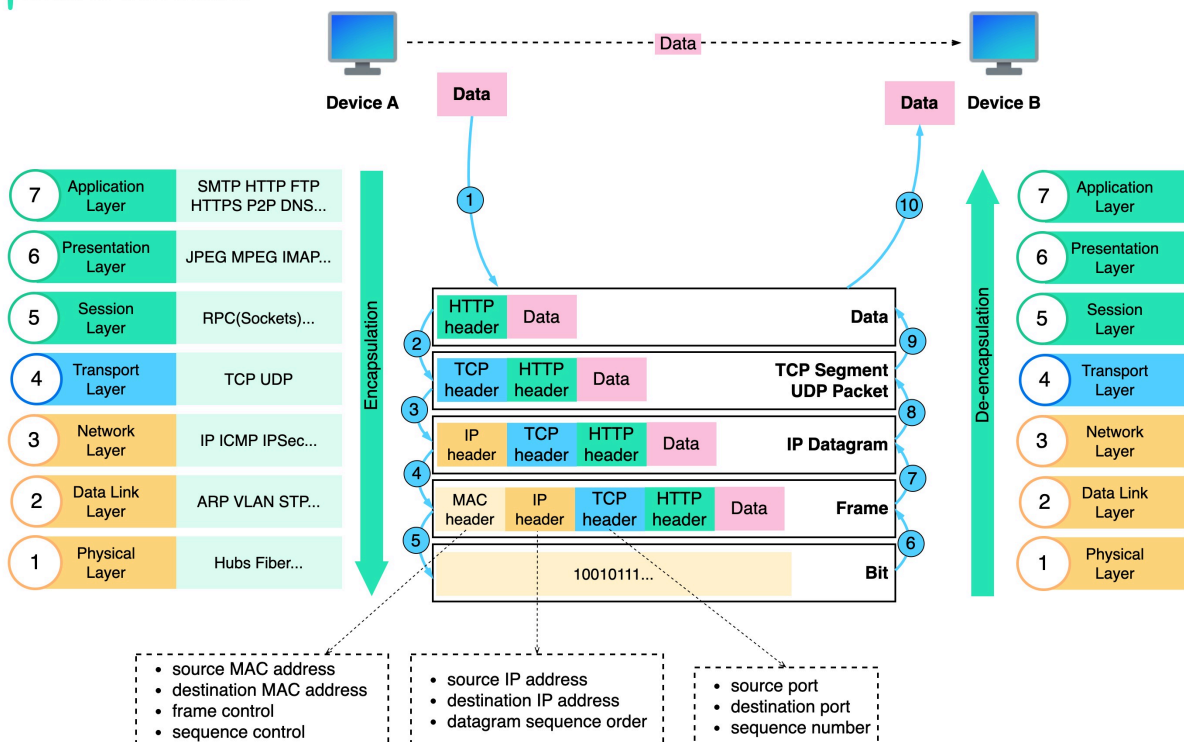
Concept	Description
LAN & WAN	Local Area Networks and Wide Area Networks.

◆ 4. OSI Model

📖 OSI Diagram

What is OSI model

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- Layer 7: Application (HTTP, DNS, FTP)
- ↓
- Layer 6: Presentation (SSL/TLS, Encoding)
- ↓
- Layer 5: Session (Session management)
- ↓
- Layer 4: Transport (TCP/UDP, Ports)
- ↓
- Layer 3: Network (IP, Routing, ICMP)
- ↓
- Layer 2: Data Link (MAC, ARP, Switching)



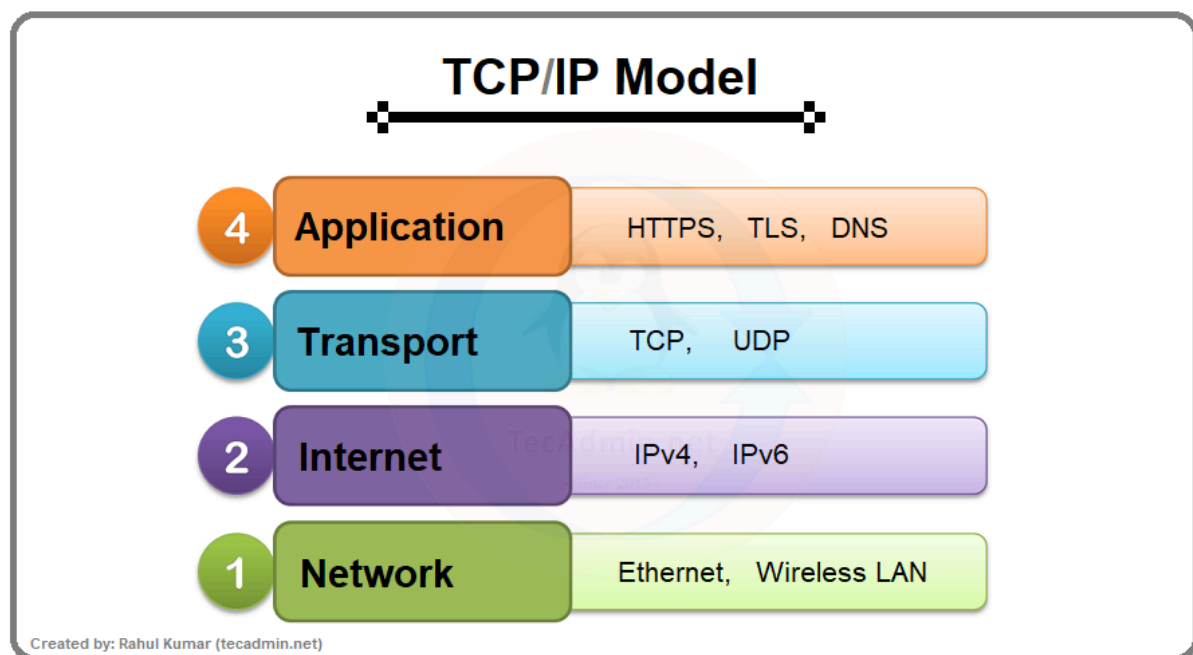
■ Layer 1: Physical (Cables, NIC, Signals)

📌 Hacker Perspective

- L7 → SQLi, XSS, SSRF
- L4 → SYN Flood, UDP Flood
- L3 → IP spoofing, ICMP tunneling
- L2 → ARP poisoning, MAC flooding
- L1 → Physical tapping

◆ 5. TCP/IP Model

📖 TCP/IP Diagram



■ Application (DNS, HTTP, HTTPS, SMTP)



■ Transport (TCP/UDP)



■ Internet (IP, ICMP, Routing)



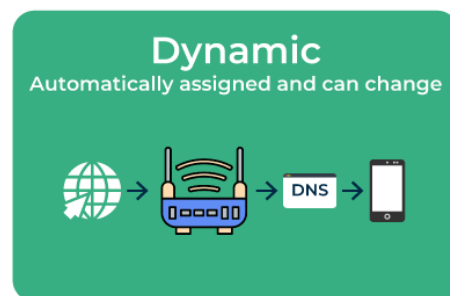
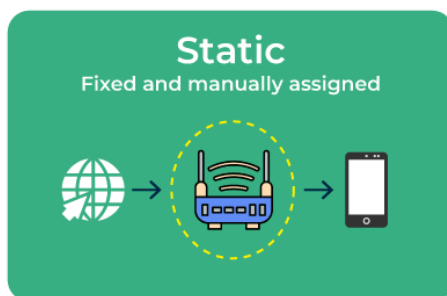
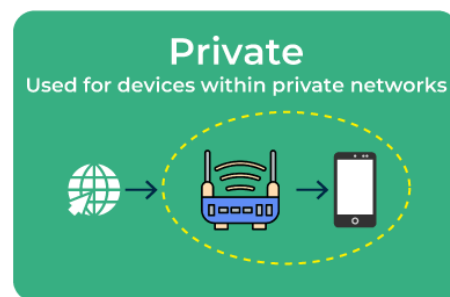
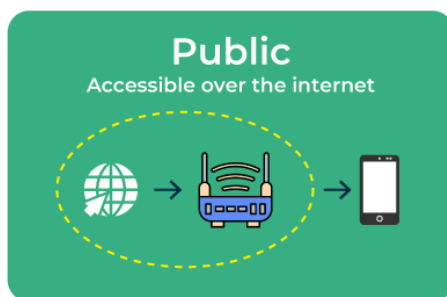
Network Access (MAC, ARP, Ethernet)

Hacker Usage

- Packet crafting → Scapy, hping3
- Spoofing → IP & ARP spoofing
- Sniffing → Wireshark, tcpdump

6. IP Addressing

Types of IP Addresses



-  **Public IP** – Internet facing
-  **Private IP** – Internal LAN
-  **Static IP** – Permanent
-  **Dynamic IP** – DHCP assigned
-  **IPv4** – 32-bit

-  **IPv6** – 128-bit

 *Hacker relevance: OSINT, network mapping, pivoting, IPv6-specific attacks*

◆ 7. MAC Address – Hardware Identity

A permanent hardware identifier used in Layer 2 communications.

 **Hacker Use:**

- ARP spoofing
 - MAC cloning
 - MITM attacks
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◆ 8. Ports – Gateway to Services

Services run on specific ports (FTP 21, SSH 22, HTTP 80, HTTPS 443, SMB 445).

 *Hackers scan ports to uncover: services, versions, misconfigs.*

Tools → **Nmap, Masscan, Netcat**

◆ 9. Server & Client Model

Understanding who *serves* and who *requests* data helps with exploitation paths.

 *Server-side vulns lead to full system compromise.*

 *Client-side vulns enable phishing & session hijack.*

◆ 10. Inbound vs Outbound Traffic

- **Inbound:** Internet → internal network
- **Outbound:** Internal → external world

 *Outbound often abused for C2 channels and data exfiltration.*

◆ 11. Subnetting

Subnetting divides a network into smaller segments.

Why hackers care?

- Helps identify target ranges
- Useful during network scans
- Allows precise reconnaissance

📖 Key elements:

Network ID → 192.168.1.0

Host Range → 192.168.1.1–192.168.1.254

Broadcast → 192.168.1.255

Mask → /24

📌 *Used for quiet scanning, internal mapping, segmentation testing.*

◆ 12. Critical Protocols for Hackers

📖 Web & Transport Protocols

Protocol(s)	Ports	Why Hackers Care	Tools
TCP / UDP	Many (Transport Layer)	TCP is connection-oriented (risks: session hijacking, SYN floods). UDP is connectionless (risks: amplification attacks, stealthy services).	<code>nmap -sT</code> / <code>nmap -sU</code> , tcpdump
HTTP / HTTPS	80 (HTTP), 443 (HTTPS)	Web applications run here → vulnerable to XSS, SQLi, SSRF, auth flaws. HTTPS encrypts data but doesn't prevent app-level attacks.	Burp Suite, curl, Nikto
DNS	53 (UDP/TCP)	Used heavily in recon → zone transfers,	dig, dnsrecon, subfinder

Protocol(s)	Ports	Why Hackers Care	Tools
		subdomain enumeration, DNS tunneling & exfiltration.	

Remote Access & File Transfer

Protocol(s)	Ports	Why Hackers Care	Tools
SSH	22	Remote shell access → risks include brute-force attacks, weak keys, misconfigured access, and lateral movement opportunities.	ssh, hydra, ssh-keyscan
FTP / SFTP	21 (FTP), 22 (SFTP)	FTP exposes credentials in plaintext and often allows anonymous access or upload. SFTP is secure, but weak credentials still pose risk.	ftp, lftp, nc

Mail & Windows Services

Protocol(s)	Ports	Why Hackers Care	Tools
SMTP / IMAP / POP3	25/587/465 (SMTP), 143/993 (IMAP), 110/995 (POP3)	Common for phishing, mail spoofing, mailbox takeover, and email-based injection attacks.	swaks, openssl s_client, telnet
SMB	445 / 139	Critical for Windows shares; vulnerable to EternalBlue, NTLM relay, and lateral movement techniques.	smbclient, enum4linux, crackmapexec
LDAP	389 / 636	Key protocol for Active Directory enumeration, user lookup, and privilege discovery.	ldapsearch, impacket

Protocol(s)	Ports	Why Hackers Care	Tools
RDP	3389	Remote Desktop access; high-value for credential theft, brute-force, persistence, and post-exploitation.	freerdp, rdpcheck

◆ 13. Network Discovery Tools

Tool	Function	Description	Example Command
Nmap	Port scanning, OS detection	The Swiss Army knife for scanning, service detection, OS fingerprinting, scripts	<code>nmap -sS -sV -A <target></code>
Masscan	High-speed port scanner	Extremely fast command-line scanner for huge IP ranges	<code>masscan 10.0.0.0/8</code>
Netcat (nc)	TCP/UDP scanner, banner grabber	Used for banner grabbing, raw connections, reverse shells	<code>nc <target> <port></code>
Subfinder	Passive recon, subdomain enumeration	Subdomain enumeration using passive sources	<code>subfinder -d <domain></code>
dnsrecon	DNS enumeration	Performs DNS recon including transfers, brute force, record enumeration	<code>dnsrecon -d <domain></code>

◆ 14. Packet Capture & Traffic Analysis Tools

Tool	Purpose	Example Usage
Wireshark	GUI-based packet analyzer for visualizing traffic, protocols, and TCP streams	Apply filters like: <code>http</code> or <code>tcp.port == 80</code>

Tool	Purpose	Example Usage
tcpdump	Command-line packet capture tool	<code>sudo tcpdump -i eth0 -w capture.pcap</code>
tshark	CLI version of Wireshark; great for automation & scripting	<code>tshark -r capture.pcap -Y "http.request"</code>
ngrep	Live traffic interceptor for searching strings inside packets	<code>ngrep "password" tcp port 80</code>

◆ 15. VPN & Proxy

Used for anonymity, bypassing restrictions, and safe testing.

VPN (Virtual Private Network):

Encrypts traffic → hides your real IP.

Proxy Servers:

Routes requests through another server; not always encrypted.

Use cases in hacking:

- Anonymous footprinting
- Bypassing IP rate-limits
- Evading geolocation restrictions

📌 *Used to avoid attribution, bypass blocks, and hide scanning origins.*